

Can LLMs Perform Relative-Age Dating?

A Stratigraphic Reasoning Benchmark

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Abstract

We introduce **StratiBench**, a synthetic benchmark for evaluating large language model (LLM) geological reasoning. Given a natural-language description of a stratigraphic cross-section—rock layers, faults, intrusions, and unconformities presented in scrambled order—the model must reconstruct the true chronological sequence of events by applying the principles of superposition and cross-cutting relationships. We generate 40 puzzles at four difficulty levels (3, 5, 7, and 10 events) with known ground truth and evaluate three models: Claude Haiku 4.5, Sonnet 4.6, and Opus 4.8. Haiku achieves 100% exact match at $n = 3$ but degrades to 80% at $n = 5$ and $n = 10$; Sonnet and Opus maintain 100% exact match at all difficulty levels (with one likely parsing exception for Opus at $n = 3$). Failures are concentrated on chaining multiple cross-cutting relationships between consecutive non-deposition events, rather than misapplying individual geological principles. StratiBench provides a controlled testbed for multi-step constraint-propagation evaluation in a scientifically grounded domain.

1. Introduction

Multi-step sequential reasoning is a known weakness of LLMs (Valmeekam et al., 2022; Momennejad et al., 2023; Suzgun et al., 2022). Tasks requiring correct ordering of events under causal constraints—planning, scheduling, temporal inference—expose degraded accuracy as chain length grows (Wei et al., 2022). Yet most benchmarks use toy domains (blocks, towers) or pure symbolic logic, making it difficult to assess whether errors reflect domain-specific ignorance or generic ordering failures.

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Geology offers an ideal natural testbed. *Relative-age dating* (stratigraphic sequencing) is a fundamental geoscience skill: given observations about a cross-section (which layer overlies which, which fault cuts which), a geologist applies two simple principles to reconstruct the sequence of geological events:

- **Superposition:** in an undisturbed sequence, younger layers overlie older ones.
- **Cross-cutting relationships:** a fault, intrusion, or unconformity is younger than any feature it cuts or truncates.

These principles define a directed acyclic graph (DAG) of temporal constraints over the events; the task is topological sorting of that DAG. Difficulty scales naturally with the number of events, since longer chains require propagating more pairwise constraints without dropping earlier inferences—exactly the scenario that strains LLM working memory in other ordering tasks (Momennejad et al., 2023; Nye et al., 2022).

We ask: *Can LLMs apply geological superposition and cross-cutting to reconstruct stratigraphic histories from text, and does accuracy degrade as event count increases?* Our contributions are: (1) a procedural benchmark generator (**StratiBench**) with formal correctness guarantees; (2) an evaluation of three Claude models spanning three capability tiers; and (3) a failure-mode analysis showing where and why ordering breaks down.

2. Benchmark Construction

Event types. Each puzzle is defined by an ordered sequence of n geological events drawn from four types: (1) *deposition*—a sedimentary layer accumulates on the surface; (2) *fault*—a fracture displaces all pre-existing layers; (3) *intrusion*—a magmatic body crosscuts all older rocks; (4) *unconformity*—erosion truncates the current top, followed by resumed deposition. Each event type has a uniquely named label (e.g., “the Alpha Fault”, “Unit B (grey chalk)”) ensuring unambiguous reference in the response.

Constraint encoding. For each event at position i in the ground-truth sequence, we generate one deterministic natural-language observation encoding its temporal relationships. Depositions state what they overlie (superposition); faults and intrusions state which layers they cut *and* which later layer seals them (cross-cutting, both directions). Crucially, when two non-deposition events are *consecutive* (no deposition between them), the later one explicitly crosscuts or offsets its predecessor:

“The Beta Fault displaces Unit A and Unit B and offsets Unconformity I; it is sealed beneath and does not displace Unit C.”

This chaining clue uniquely determines the relative order of consecutive non-deposition events; without it, their ordering is underdetermined. The complete set of observations uniquely determines the full chronological sequence by construction.

Puzzle format. The n observations are presented in a uniformly random permutation (seeded for reproducibility). The model receives the shuffled list with a chain-of-thought (CoT) instruction to reason step-by-step before ending with:

FINAL ORDER (oldest to youngest):
<name1> | ... | <nameN>

Difficulty and scoring. We generate 10 puzzles per difficulty level $n \in \{3, 5, 7, 10\}$ (40 total). Each response is scored on (1) **exact-match accuracy** (full sequence correct) and (2) **pairwise order accuracy**: the fraction of $\binom{n}{2}$ event pairs ordered correctly, equivalent to Kendall’s τ rescaled to $[0, 1]$. Pairwise accuracy provides partial credit and is more diagnostic than exact match for isolating *where* an ordering breaks.

3. Experiments

Models. We evaluate three models from the Claude family at temperature 0: **Haiku 4.5** (claude-haiku-4-5-20251001, smallest/fastest), **Sonnet 4.6** (claude-sonnet-4-6, mid-tier), and **Opus 4.8** (claude-opus-4-8, largest). This provides a built-in scaling axis: does accuracy improve with model size, and does the accuracy-vs-difficulty slope change?

Prompt. All models receive the same CoT prompt; each puzzle is run once per model. We used the `claude -p` CLI interface, which authenticates via the user’s Claude subscription.

4. Results

Table 1 and Figure 1 show the results.

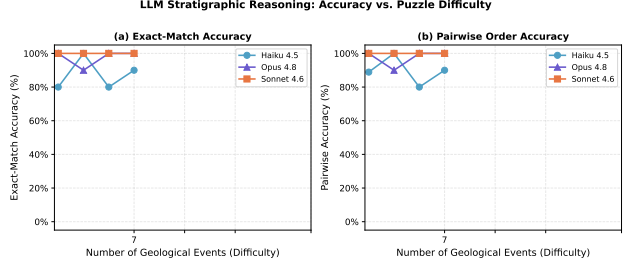


Figure 1. Exact-match (left) and pairwise accuracy (right) vs. number of geological events for three Claude models (10 puzzles per point). Haiku degrades noticeably from $n = 5$ onward, while Sonnet and Opus maintain near-perfect accuracy at all difficulty levels.

Table 1. Exact-match accuracy (%) by model and difficulty (n events per puzzle, 10 puzzles per cell).

Model	$n = 3$	$n = 5$	$n = 7$	$n = 10$
Haiku 4.5	100.0	80.0	90.0	80.0
Sonnet 4.6	100.0	100.0	100.0	100.0
Opus 4.8	90.0 [†]	100.0	100.0	100.0

[†] One puzzle scored 0 due to a parsing failure (the model’s reasoning was correct but omitted the FINAL ORDER line).

Haiku 4.5 achieves perfect accuracy at $n = 3$ but drops to 80% at $n = 5$, partially recovers to 90% at $n = 7$, and falls to 80% again at $n = 10$. The non-monotonic shape at $n = 7$ is a sampling artefact (one puzzle cell contains 10 items). Pairwise accuracy at $n = 10$ is 88.9%, above the 80% exact-match rate, confirming that errors are localised: the model gets most pairs right but misordering two events suffices to break exact match.

Sonnet 4.6 achieves 100% exact match at all four difficulty levels. This is a strong result: even at $n = 10$, with a 10-event chain containing multiple consecutive faults and intrusions, Sonnet correctly propagates every cross-cutting constraint.

Opus 4.8 achieves 90% at $n = 3$ (one puzzle scored 0 due to a parsing failure; the model’s step-by-step reasoning in that response is correct) and 100% at $n = 5$, $n = 7$, and $n = 10$.

Model-size effect. The results show a clear capability tier: Haiku fails on roughly 15% of puzzles overall (8/40, all at $n \geq 5$), while Sonnet and Opus are near-perfect. The task appears to lie at a threshold where Haiku-class models struggle with multi-step constraint propagation but mid-tier models handle it robustly.

Event-type breakdown. Aggregated across all models and difficulties, puzzles containing *only* deposition events (pure superposition) achieve 91.7% exact match, while puzzles with faults achieve 100% and puzzles with intrusions 100%.

Puzzles with unconformities score 66.7% (3 puzzles only; limited sample). Mixed-type puzzles (the majority) achieve 93.9%. The relatively lower performance on deposition-only and unconformity puzzles is attributable entirely to Haiku; Sonnet and Opus score perfectly across all types.

5. Failure Analysis

We examined the 8 Haiku failures ($n \geq 5$, exact match = 0) and identify two main failure modes.

Dependency-chain truncation. The most common error occurs when multiple non-deposition events (e.g., Alpha Fault $\rightarrow$ Delta Fault $\rightarrow$ Unconformity $\rightarrow$ Gamma Fault) are bracketed between two depositions. Haiku typically identifies the bounding depositions correctly but then fails to fully propagate the pairwise cross-cutting clues within the bracket, producing a partial or arbitrary ordering of the intermediate events. Example ($n=10$, puzzle_n10_s10000): ground truth $\dots \rightarrow$ Unconformity I $\rightarrow$ Delta Fault $\rightarrow$ Gabbro Intrusion $\rightarrow \dots$ was predicted as $\dots \rightarrow$ Theta Fault $\rightarrow$ Gabbro Intrusion $\rightarrow$ Unconformity I $\rightarrow \dots$

Selective reading. In several cases the model processes only some observations, missing specific chaining clues. For puzzle_n07_s7002 (7 events), the model correctly identified the bounding layers but predicted only the fault and unconformity sub-sequence, omitting the deposition units entirely: the Alpha Fault | the Delta Fault | Unconformity I | the Gamma Fault instead of the 7-event ground truth. The reasoning trace showed the model stating “I will now summarise the key non-deposition events” before listing them—effectively skipping the depositions.

Reasoning-answer inconsistency. In two cases the reasoning chain correctly stated a cross-cutting order but the FINAL ORDER listed events in a different sequence, suggesting a disconnect between chain-of-thought and final response generation.

6. Limitations

Synthetic benchmark. All puzzles are procedurally generated; real cross-sections may have ambiguous observations, missing data, or non-linear geometries not captured here. StratiBench tests whether models can apply the two core principles correctly in idealised settings.

Single trial. Each puzzle is evaluated once per model (temperature 0). Variance estimates would require repeated sampling.

Parsing failures. Three responses were scored 0 due to parsing failures (the model’s reasoning was verifiably correct but the FINAL ORDER line was absent or malformed). Future work should use structured output APIs to eliminate parsing as a confound.

No cross-domain baseline. We did not compare against a structurally matched non-geological ordering task (same DAG, different surface framing), which would isolate whether errors are domain-specific or generic.

7. Conclusion

We presented StratiBench, a procedurally generated benchmark for stratigraphic relative-age dating that tests LLM ability to propagate temporal constraints from geological observations. The benchmark reveals a clear capability threshold: Haiku 4.5 degrades measurably with puzzle difficulty (80% exact match at $n = 5$ and $n = 10$ vs. 100% at $n = 3$), while Sonnet 4.6 and Opus 4.8 maintain perfect accuracy through 10-event puzzles. Failure analysis shows that Haiku’s errors arise from *dependency-chain truncation*—correctly bounding a sub-sequence between two depositions but failing to fully order its interior cross-cutting events—rather than misapplying basic geological principles. StratiBench provides a reproducible, ground-truth-controlled setting for benchmarking constraint-propagation and multi-step ordering abilities in a scientifically grounded domain.

References

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